

A first-order differential equation

A first-order differential equation relates a variable, its time derivative, and possibly time:

$$\dot{z}_t = g(z_t, t), \quad \dot{z}_t \equiv dz_t/dt.$$

A **linear** first-order equation has the form

$$\dot{z}_t + a(t)z_t = b(t).$$

With an initial condition z_0 , it determines the path z_t . We begin with constant coefficients $a > 0$ and b .

Solve the linear equation: integrating factor

Consider $\dot{z}_t + az_t = b$, where $a > 0$. Multiply both sides by e^{at} :

$$e^{at}\dot{z}_t + ae^{at}z_t = be^{at}.$$

The product rule turns the left-hand side into a single derivative:

$$\frac{d}{dt}(e^{at}z_t) = be^{at}.$$

Integrate both sides with respect to time:

$$\int \frac{d}{dt}(e^{at}z_t) dt = \int be^{at} dt \implies e^{at}z_t = \frac{b}{a}e^{at} + C,$$

where C is an arbitrary constant of integration.

Use the initial condition to determine the constant

Dividing by e^{at} gives the family of solutions

$$z_t = \frac{b}{a} + Ce^{-at}.$$

At $t = 0$, $e^{-a \cdot 0} = 1$, so the initial condition $z(0) = z_0$ implies

$$z_0 = \frac{b}{a} + C \quad \implies \quad C = z_0 - \frac{b}{a}.$$

Substituting this value of C yields the particular solution

$$z_t = \frac{b}{a} + \left(z_0 - \frac{b}{a} \right) e^{-at}.$$

Interpret the linear solution

The stationary value is $z^* = b/a$, so the solution becomes

$$z_t = z^* + (z_0 - z^*)e^{-at}.$$

- ▶ The gap $z_t - z^*$ shrinks at exponential rate a . Differentiating with respect to time,

$$\frac{d}{dt}(z_t - z^*) = -a(z_0 - z^*)e^{-at} = -a(z_t - z^*).$$

- ▶ If $z_0 < z^*$, the path rises toward z^* ; if $z_0 > z^*$, it falls.
- ▶ At time t , the fraction of the initial gap closed is $1 - e^{-at}$.

The Solow equation is nonlinear

With Cobb–Douglas production per effective worker, $\hat{y}_t = \hat{k}_t^\alpha$,

$$\dot{\hat{k}}_t = s\hat{k}_t^\alpha - (\delta + n + x)\hat{k}_t, \quad 0 < \alpha < 1.$$

The term $\hat{k}_t^\alpha$ makes this equation nonlinear in $\hat{k}_t$. Its positive steady state is

$$\hat{k}^* = \left(\frac{s}{\delta + n + x} \right)^{1/(1-\alpha)}.$$

The Cobb–Douglas equation is a Bernoulli equation: a change of variable makes it linear.

Transform the Solow equation

For $\hat{k}_t > 0$, divide the equation by $\hat{k}_t^\alpha$:

$$\hat{k}_t^{-\alpha} \dot{\hat{k}}_t + (\delta + n + x) \hat{k}_t^{1-\alpha} = s.$$

Define $z_t \equiv \hat{k}_t^{1-\alpha}$. By the chain rule,

$$\dot{z}_t = (1 - \alpha) \hat{k}_t^{-\alpha} \dot{\hat{k}}_t.$$

Consequently,

$$\dot{z}_t + (1 - \alpha)(\delta + n + x)z_t = (1 - \alpha)s.$$

This is linear in z_t .

The exact Solow transition path

The transformed equation is linear:

$$\dot{z}_t + (1 - \alpha)(\delta + n + x)z_t = (1 - \alpha)s.$$

Its steady-state value is

$$z^* \equiv \frac{s}{\delta + n + x} = (\hat{k}^*)^{1-\alpha}.$$

Applying the linear solution gives

$$z_t = z^* + (z_0 - z^*)e^{-\beta t}, \quad \beta \equiv (1 - \alpha)(\delta + n + x).$$

Since $z_t = \hat{k}_t^{1-\alpha}$, the exact path for capital is

$$\hat{k}_t = \left[z^* + (\hat{k}_0^{1-\alpha} - z^*)e^{-\beta t} \right]^{1/(1-\alpha)}.$$

Exact and approximate convergence

The exact solution says the gap in $z_t = \hat{k}_t^{1-\alpha}$ closes at rate

$$\beta = (1 - \alpha)(\delta + n + x).$$

The path for $\hat{k}_t$ itself is nonlinear. Near the steady state, however, its *logarithm* approximately follows the same exponential convergence law.

Why approximate?

Log output per effective worker is proportional to log capital per effective worker. This makes the approximation useful for measuring income convergence.

Log-linearize capital around the steady state

Set $\tilde{k}_t \equiv \ln \hat{k}_t$. The Solow equation implies

$$\dot{\tilde{k}}_t = \frac{\dot{\hat{k}}_t}{\hat{k}_t} = s e^{-(1-\alpha)\tilde{k}_t} - (\delta + n + x) \equiv h(\tilde{k}_t).$$

A first-order Taylor expansion of h around $\tilde{k}^* = \ln \hat{k}^*$ is

$$\dot{\tilde{k}}_t = h(\tilde{k}_t) \simeq \underbrace{h(\tilde{k}^*)}_{=\dot{\tilde{k}}^*=0} + h'(\tilde{k}^*)(\tilde{k}_t - \tilde{k}^*).$$

Thus we need the slope $h'(\tilde{k}^*)$ to describe how the deviation from steady state evolves.

Evaluate the slope at the steady state

Differentiate the function on the previous slide:

$$h'(\tilde{k}) = -(1 - \alpha)se^{-(1-\alpha)\tilde{k}}.$$

At $\tilde{k}^*$, the steady-state condition implies

$$se^{-(1-\alpha)\tilde{k}^*} = \delta + n + x.$$

Hence

$$h'(\tilde{k}^*) = -(1 - \alpha)(\delta + n + x) \equiv -\beta.$$

Substitution into the Taylor approximation gives the linear equation

$$\dot{\tilde{k}}_t \simeq -\beta(\tilde{k}_t - \tilde{k}^*), \quad \beta = (1 - \alpha)(\delta + n + x).$$

Solve the log-linearized equation

Rearrange the approximate equation:

$$\dot{\tilde{k}}_t + \beta \tilde{k}_t = \beta \tilde{k}^*.$$

Compare it with the linear equation solved earlier, $\dot{z}_t + az_t = b$:

$$z_t \longleftrightarrow \tilde{k}_t, \quad a \longleftrightarrow \beta, \quad b \longleftrightarrow \beta \tilde{k}^*.$$

Its stationary value b/a is therefore $\tilde{k}^*$. Applying $z_t = b/a + (z_0 - b/a)e^{-at}$ gives

$$\tilde{k}_t \simeq \tilde{k}^* + (\tilde{k}_0 - \tilde{k}^*)e^{-\beta t}.$$

Equivalently, $\tilde{k}_t - \tilde{k}_0 \simeq (1 - e^{-\beta t})(\tilde{k}^* - \tilde{k}_0)$.

Translate capital convergence into income convergence

Let $y_t \equiv Y_t/L_t = A_t \hat{y}_t$ and $\hat{y}_t = \hat{k}_t^\alpha$. Then

$$\ln y_t = \ln A_t + \alpha \tilde{k}_t, \quad \ln \hat{y}^* = \alpha \tilde{k}^*.$$

For constant technology growth x , $\ln A_t - \ln A_0 = xt$. Therefore,

$$\ln y_t - \ln y_0 \simeq xt + (1 - e^{-\beta t}) [\ln \hat{y}^* - (\ln y_0 - \ln A_0)].$$

Conditional on the determinants of $\hat{y}^*$, a lower initial output level predicts faster transitional growth. The approximation applies near the steady state.

From the Solow transition to a growth regression

Let i index countries and T denote the length of the observation interval. Define

$$\lambda_T \equiv \frac{1 - e^{-\beta T}}{T}.$$

Substituting the Cobb–Douglas steady state into the transition equation gives

$$\frac{\ln y_{i,T} - \ln y_{i,0}}{T} \simeq c_i - \lambda_T \ln y_{i,0} + \lambda_T \frac{\alpha}{1 - \alpha} \ln s_i - \lambda_T \frac{\alpha}{1 - \alpha} \ln(n_i + x + \delta),$$

where

$$c_i \equiv x + \lambda_T \ln A_{i,0}.$$

For estimation, initial productivity may be related to observed country characteristics $\mathbf{Z}_i$:

$$c_i = c_0 + \gamma' \mathbf{Z}_i + u_i.$$

Thus $\mathbf{Z}_i$ can include measures such as rule of law, life expectancy, and inflation.

Conditional convergence predicts a negative coefficient on initial income, holding the determinants of each country's steady state fixed.

The speed of convergence and half-life

In the basic model, $\beta = (1 - \alpha)(\delta + n + x)$. If $\alpha = 1/3$, $\delta = 0.05$, $n = 0.01$, and $x = 0.02$ per year, then

$$\beta = \frac{2}{3}(0.08) \simeq 0.0533 \quad \text{per year.}$$

The time required to close half the initial log gap solves $e^{-\beta T_{1/2}} = 1/2$:

$$T_{1/2} = -\frac{\ln(1/2)}{\beta}.$$

At $\beta = 0.0533$, the half-life is about 13 years; at an illustrative empirical rate $\beta = 0.02$, it is about 35 years. A slower rate motivates richer models of convergence.